

Exponential functions - 0.1.5

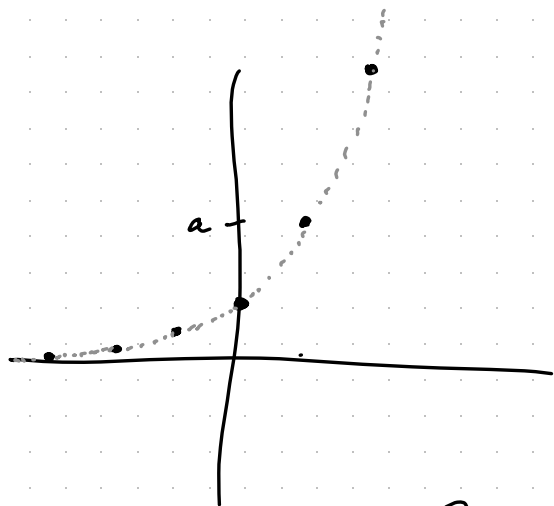
Let $a > 0$. Know what a^3 is: it's $a \cdot a \cdot a$.
generally, a^n is $\underbrace{a \cdot a \cdot \dots \cdot a}_{n \text{ times}}$, and we can
define $a^0 = 1$.

For negative integers, $a^{-n} = \frac{1}{a^n}$

($a^{-n} \cdot a^n = a^{-n+n} = a^0 = 1$, so want $a^{-n} = \frac{1}{a^n}$)

Goal: define a function a^x where the
exponent is the input.

Problem: this means we need to know
what a^π is, or $a^{\sqrt{2}}$, e.g.



The function x^n is
a 1-1 function when
the domain is $[0, \infty)$
which means it has an
inverse $\sqrt[n]{x}$.

Define $a^{\frac{1}{n}}$ to be $\sqrt[n]{a}$ and
define $a^{\frac{m}{n}}$ to be $(\sqrt[n]{a})^m$.

$$\left(a^{\frac{m}{n}} = (a^{\frac{1}{n}})^m = (\sqrt[n]{a})^m \right).$$

Theorem: If we know what f does to
all rational inputs, then there is at
most one continuous function defined on all
real numbers and agreeing with f .

Using the theorem, there is a unique continuous way to extend $a^{\frac{m}{n}}$ to a function that takes real inputs, call it a^x .

To compute a^{π} , find the limit of the sequence $a^3, a^{3.1}, a^{3.14}, a^{3.141}, a^{3.1415}, \dots$

If $a=2$, we can compute $2^{3.14} \approx 8.82$

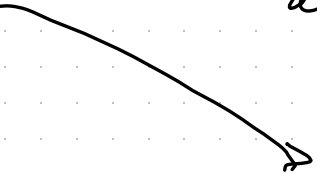
Point: this sequence is something we can compute, and 2^{π} is the limit value.

We have: $x < y$ implies that $a^x < a^y$.

Properties of a^x :

- increasing, hence one-to-one and invertible
- $a^x \cdot a^y = a^{x+y}$
- $\frac{a^x}{a^y} = a^{x-y}$
- $(a^x)^y = a^{xy}$
- $(ab)^x = a^x b^x$
- $\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$

even when x and y
are irrational.


$$(a^n)^{\frac{1}{n}} = a^{n \cdot \frac{1}{n}} = a^1 = a$$